

Estimation Theory

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REVIEW OF PROBABILITY

AXIOMS, CONDITIONAL PROBABILITY, INDEPENDENCE

Probability Space

A **probability space** is a triple $(\Omega, \mathcal{F}, P)$

- Ω is a set of outcomes
- $\mathcal{F}$ is a set of events
- $P: \mathcal{F} \rightarrow [0,1]$ is a function that assigns probabilities to events

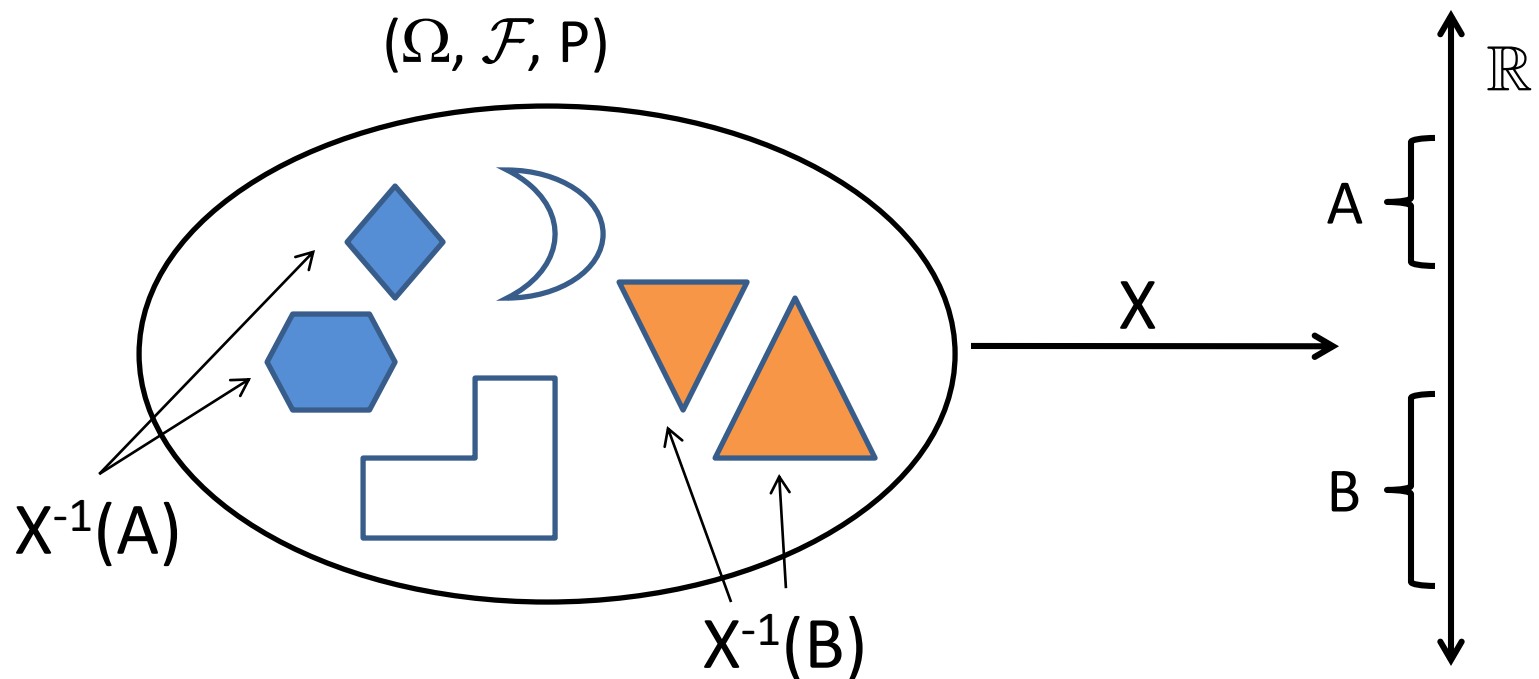
Note: $\mathcal{F}$ is a σ -field, i.e., collection of subsets of Ω such that

- If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$
- If $A_i \in \mathcal{F}$ is a countable sequence of sets then $\bigcup_i A_i \in \mathcal{F}$

Random Variables (r.v.)

- $X: \Omega \rightarrow \mathbb{R}$ is a **random variable** if for every measurable set $B \subset \mathbb{R}$,

$$X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F} .$$



Probability Axioms

1. Nonnegativity

$$P(A) \geq 0 \quad \forall A \in \mathcal{F}$$

2. Normalization

$$P(\Omega) = 1$$

3. Additivity

$A_1, A_2, \dots$ disjoint events

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$$

Conditional Probability

Conditional probability

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Multiplication rule

$$P(B \cap A) = P(A)P(B|A)$$

Bayes rule

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

Example

A: blue on first draw

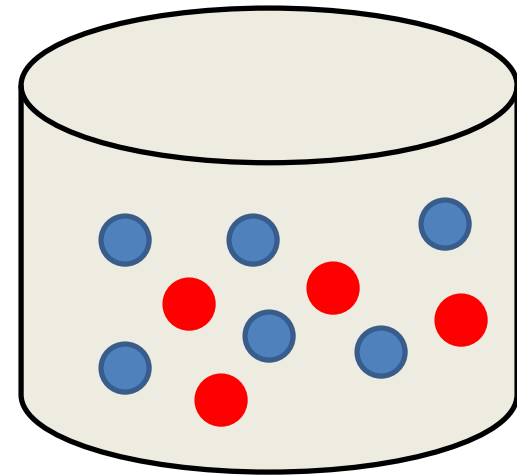
B: blue on second draw

$P(A, B) = ?$

$$P(A) = \frac{6}{10}$$

$$P(B|A) = \frac{5}{9}$$

$$P(A, B) = \frac{1}{3}$$



Urn

Independence

Events A and B are **independent** if $P(B|A) = P(B)$,
i.e., $P(A \cap B) = P(A) P(B)$.

Example

Roll 2 dice

A = 1st die is 4

B_1 = 2nd die is 2

B_2 = sum of 2 dice is 9

B_3 = sum of 2 dice is 7



Q1: Are A and B_1 independent?

Q2: Are A and B_2 independent?

Q3: Are A and B_3 independent?

Some Discrete Random Variables

- **Bernoulli distribution:** $X \sim \text{Bernoulli}(p)$

Bernoulli 2
n=1 vs 0

$$\begin{aligned}P(X = 1) &= p \\P(X = 0) &= 1 - p\end{aligned}$$

"Coin tossing"

- **Binomial distribution:** $X \sim \text{Binomial}(n, p)$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k = 0, \dots, n$$

- **Geometric distribution:** $X \sim \text{Geometric}(p)$

$$P(X = n) = (1 - p)^{n-1} p \quad \text{for } n = 1, 2, \dots$$

- **Poisson distribution:** $X \sim \text{Poisson}(\lambda)$, where $\lambda > 0$

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

Density, Distribution

Density function (f)

$$\forall x \quad f(x) \geq 0 \quad \int_{-\infty}^{\infty} f(x)dx = 1 \quad P(a \leq X \leq b) = \int_a^b f(x)dx$$

Distribution function (F)

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= P(-\infty \leq X \leq x) \\ &= \int_{-\infty}^x f(x)dx \end{aligned}$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

Exponential Distribution

$$f(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

When $T \sim \text{Exp}(\lambda)$, what is $P(T > t + s | T > t)$?

$$\begin{aligned} P(T > t) &= 1 - P(T \leq t) \\ &= 1 - (1 - e^{-\lambda t}) \\ &= e^{-\lambda t} \end{aligned}$$

$$\begin{aligned} P(T > t + s, T > t) &= P(T > t + s) \\ &= e^{-\lambda(t+s)} \end{aligned}$$

$$\begin{aligned} P(T > t + s | T > t) &= \frac{P(T > t + s, T > t)}{P(T > t)} \\ &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} \\ &= e^{-\lambda s} \\ &= P(T > s) \end{aligned}$$

Memoryless property of the exponential distribution

Joint Distribution, Marginal Distribution

- Two random variables X and Y .
- **Joint** distribution:

$$P(X, Y)$$

- **Marginal** distribution:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

- If X and Y are independent,

$$\begin{aligned} P(X + Y = z) &= \sum_x P(X = x, Y = z - x) \\ &= \sum_x P(X = x)P(Y = z - x) \end{aligned}$$

EXPECTED VALUE, MOMENTS

Expected Value

X discrete r.v.

$$\mathbb{E}h(X) = \sum_x h(x)P(X = x)$$

$h(x) = x$	$\mathbb{E}X$	expected value of X
$h(x) = x^k$	$\mathbb{E}X^k$	k -th moment of X
$h(X) = (X - \mathbb{E}(X))^2$	$\mathbf{var}(X)$	variance of X

Standard deviation: $\sigma(X) = \sqrt{\mathbf{var}(X)}$.

Poisson Distribution

For $k = 0, 1, 2, \dots$,

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

$$\begin{aligned}\mathbb{E}(X) &= \sum_{k=0}^{\infty} kP(X = k) \\ &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \lambda \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} = \lambda \sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \\ &= \lambda \left(\sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \right) \\ &= \lambda\end{aligned}$$

Practice: $\text{var}(X) = \lambda$

Some Facts about Expectation

- $X_1, X_2, \dots, X_n$, random variables
- Linearity of expectation

$$\mathbb{E}(X_1 + \dots + X_n) = \mathbb{E}X_1 + \dots + \mathbb{E}X_n$$

- If $X_1, X_2, \dots, X_n$ are independent,

$$\begin{aligned}\mathbb{E}(X_1 \cdots X_n) &= \mathbb{E}X_1 \cdots \mathbb{E}X_n \\ \mathbf{var}(X_1 + \dots + X_n) &= \mathbf{var}(X_1) + \dots + \mathbf{var}(X_n)\end{aligned}$$

- For $c \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}(X + c) &= \mathbb{E}(X) + c \\ \mathbf{var}(X + c) &= \mathbf{var}(X) \\ \mathbb{E}(cX) &= c\mathbb{E}(X) \\ \mathbf{var}(cX) &= c^2\mathbf{var}(X)\end{aligned}$$

Gamma Distribution (n,λ)

$$f(x) = \frac{\lambda^n x^{n-1}}{(n-1)!} e^{-\lambda x}$$

Fact: If $t_1, t_2, \dots, t_n$ are independent and has $\text{Exp}(\lambda)$, then $T_n = t_1 + \dots + t_n$ has a gamma distribution (n, λ) .

$$\mathbb{E}T_n = \mathbb{E}(t_1 + \dots + t_n) = \mathbb{E}t_1 + \dots + \mathbb{E}t_n = n \cdot \frac{1}{\lambda}$$

$$\mathbf{var}(T_n) = \mathbf{var}(t_1 + \dots + t_n) = \mathbf{var}(t_1) + \dots + \mathbf{var}(t_n) = n \cdot \frac{1}{\lambda^2}$$

Moment-Generating Function

$$\varphi_X(t) = \mathbb{E}(e^{tX})$$

$$\varphi'_X(t) = \mathbb{E}(Xe^{tX}) \quad \varphi'_X(0) = \mathbb{E}(X)$$

$$\varphi''_X(t) = \mathbb{E}(X^2e^{tX}) \quad \varphi''_X(0) = \mathbb{E}(X^2)$$

$$\vdots$$
$$\vdots$$

$$\varphi_X^{(k)}(t) = \mathbb{E}(X^k e^{tX}) \quad \varphi_X^{(k)}(0) = \mathbb{E}(X^k)$$

Theorem: X and Y are independent. Then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

$$\begin{aligned}\varphi_{X+Y}(t) &= \mathbb{E}\left(e^{t(X+Y)}\right) \\ &= \mathbb{E}\left(e^{tX}e^{tY}\right) \\ &= \mathbb{E}\left(e^{tX}\right)\mathbb{E}\left(e^{tY}\right) \\ &= \varphi_X(t)\varphi_Y(t)\end{aligned}$$

Important property of MGF: it determines the distribution of the r.v.

$X \sim \text{Poisson}(\lambda)$. What is $\varphi_X(t)$?

$$\begin{aligned}\varphi_X(t) &= \mathbb{E}(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} e^{-\lambda} \frac{\lambda^k}{k!} \\&= e^{-\lambda} \sum_{k=0}^{\infty} \frac{e^{tk} \lambda^k}{k!} \\&= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(e^t \lambda)^k}{k!} \\&= e^{-\lambda} \exp(e^t \lambda) \\&= \exp(e^t \lambda - \lambda) = \exp(\lambda(e^t - 1))\end{aligned}$$

$X \sim \text{Poisson}(\lambda)$, $Y \sim \text{Poisson}(\mu)$, and X and Y are independent.
What is $\varphi_{X+Y}(t)$?

$$\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t)$$

$$= \exp(\lambda(e^t - 1)) \exp(\mu(e^t - 1))$$

$$= \exp((\lambda + \mu)(e^t - 1))$$

$$X + Y \sim \text{Poisson}(\lambda + \mu)$$

Laws of Large Numbers

Strong law of large numbers:

- $X_1, X_2, \dots$, i.i.d. (independently and identically distributed) with $E(X_i) = \mu$ and integrable.
- $S_n = X_1 + \dots + X_n$
- Then w.p. (with probability) one

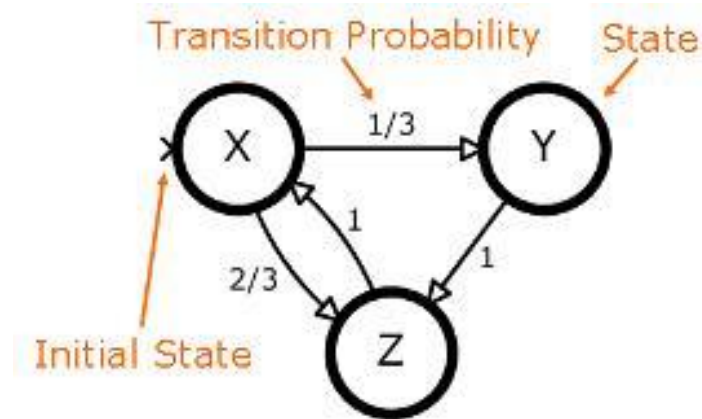
$$\frac{S_n}{n} \rightarrow \mu \text{ as } n \rightarrow \infty.$$

Central limit theorem:

- Sum of a sufficiently large number of independent random variables, each with finite mean and variance, will be approximately **normally** distributed.

Markov Chains

- An important tool in probability



- Markov property
- Classification of states
- Limit behavior, convergence, stationary distribution
- Applications: Hidden Markov model, Markov random field, Markov chain Monte Carlo, ...